

KSS report - toy

1 gene(s), 42 scored positions. Weights: discriminative 1, mutational 1, protein 1.

KSS = weighted mean of three components, each in [0, 1]:

- Discriminative - class separation, from information theory (normalized mutual information × k-mer purity).
- Mutational - biophysical impact of the amino acid changes, from the optimized MIYATA_EVO substitution matrix.
- Protein - depth of UniProt characterization, a weighted sum of ten annotation-evidence categories.

Top 12 positions by KSS (all variants)

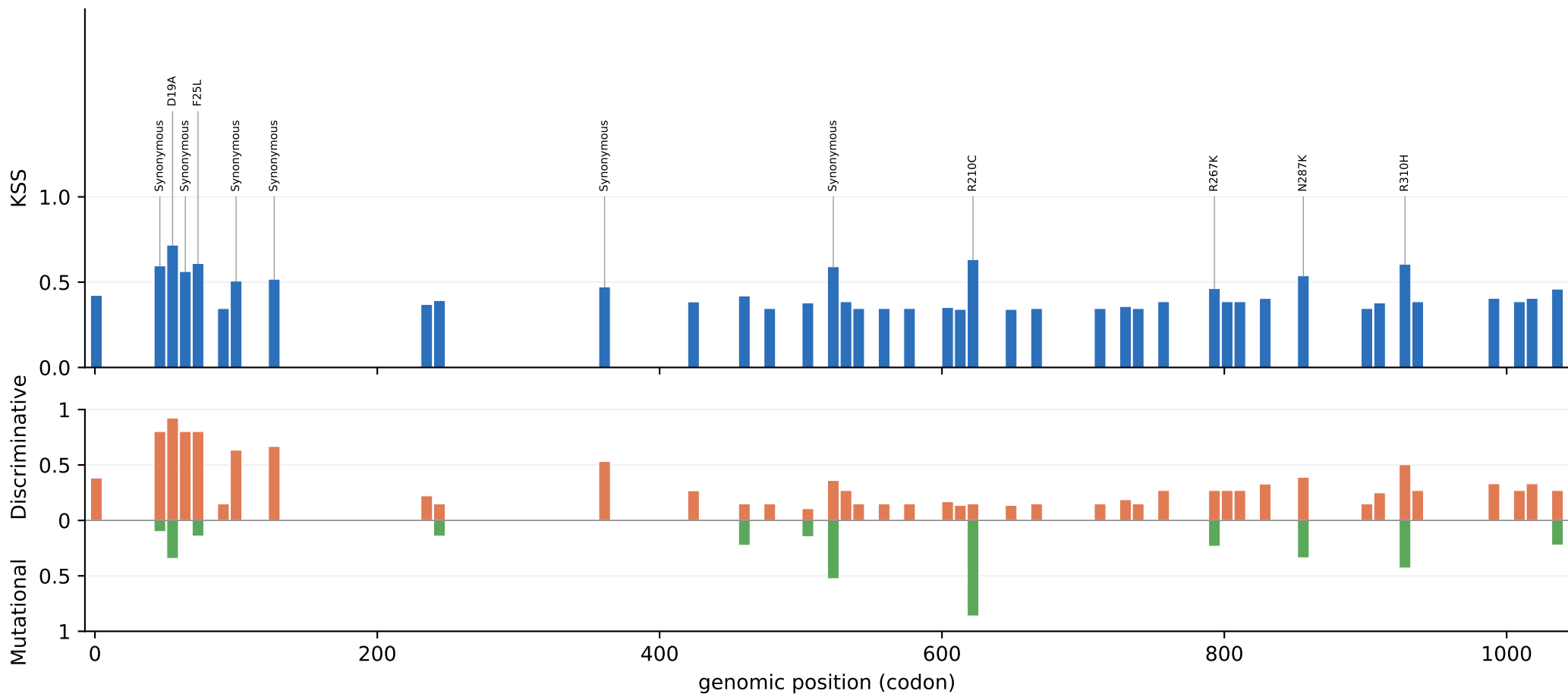
Rank	Gene	Pos	KSS	Disc	Mut	Prot	Ref k-mer	Variant	AA change	Top class
1	US28	55	0.717	0.923	0.342	0.885	GACGCGACT	GCCGCGACT	D19A (0.34)	A1 (100%)
								GACGCGGCT	T21A (0.12)	C (100%)
								GAAGCGACT	D19E (0.10)	D (100%)
								GACGCGACT	reference	A2 (100%)
2	US28	622	0.632	0.149	0.861	0.885	TACTACCGC	TACTACTGC	R210C (0.86)	A1 (100%)
								TACTACCGC	reference	A2 (27%), C (27%), D (27%), +1 others
3	US28	73	0.609	0.801	0.141	0.885	TTCACCGAC	CTAACTGAC	F25L (0.14)	C (100%)
								CTCACCGAC	F25L (0.14)	D (100%)
								TTCACCGAC	reference	A1 (50%), A2 (50%)
4	US28	928	0.605	0.502	0.429	0.885	CGCCAGCGA	CACCAGCGA	R310H (0.43)	D (75%), A2 (25%)
								CGTCAGCGA	Synonymous	C (100%)
								CGCCAGCGA	reference	A1 (40%), A2 (30%), C (20%), +1 others
5	US28	46	0.595	0.801	0.100	0.885	TACGATGAA	TATGATGAA	Synonymous	C (100%)
								TACGACGAT	E18D (0.10)	D (100%)
								TACGATGAA	reference	A1 (50%), A2 (50%)
6	US28	523	0.590	0.360	0.525	0.885	ACCGACTAC	ACTGACTAC	Synonymous	C (100%)
								ACCCACTAC	D176H (0.53)	A1 (100%)
								ACCGACTAC	reference	A2 (31%), D (31%), A1 (23%), +1 others

KSS report - toy (top positions, continued)

Top positions by KSS (continued)

Rank	Gene	Pos	KSS	Disc	Mut	Prot	Ref k-mer	Variant	AA change	Top class
7	US28	64	0.562	0.801	0.000	0.885	CCTTGTGTT	CCCTGTGTT	Synonymous	C (100%)
								CCCTGTGTC	Synonymous	D (100%)
								CCTTGTGTT	reference	A1 (50%), A2 (50%)
8	US28	856	0.537	0.389	0.336	0.885	CTCAATCCG	CTCAAAACCG	N287K (0.34)	D (100%)
								CTCAATCCG	reference	A1 (31%), A2 (31%), C (31%), +1 others
9	US28	127	0.517	0.666	0.000	0.885	GTCTTTCTC	GTCTTTCTT	Synonymous	D (100%)
								GTCTTTCTG	Synonymous	A1 (100%)
								GTCTTTCTC	reference	A2 (40%), C (40%), A1 (20%)
10	US28	100	0.506	0.633	0.000	0.885	CCAGTTACG	CCAGTCACG	Synonymous	C (44%), D (33%), A2 (22%)
								CCGGTCACG	Synonymous	A1 (67%), D (33%)
								CCGGTTACG	Synonymous	A1 (100%)
								CCAGTTACG	reference	A2 (100%)
11	US28	361	0.472	0.531	0.000	0.885	TTTATCACG	TTTATAACG	Synonymous	A2 (100%)
								TTTATCACA	Synonymous	C (100%)
								TTTATTACG	Synonymous	A1 (100%)
								TTTATCACG	reference	D (36%), A1 (27%), A2 (18%), +1 others
12	US28	793	0.462	0.269	0.232	0.885	TTCGAAAGA	TTCGAAAAA	R267K (0.23)	D (60%), A2 (20%), C (20%)
								TTCGAAAGA	reference	A1 (36%), A2 (27%), C (27%), +1 others

US28 - scores per position



G-protein coupled receptor homolog US28 | protein characterization score 0.885

UniProt P69332: <https://www.uniprot.org/uniprotkb/P69332>

GenBank reference FJ527563: <https://www.ncbi.nlm.nih.gov/nuccore/FJ527563>